

Unit 13, Lesson 4: Generating Equivalent Algebraic Expressions



Benchmark

MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.

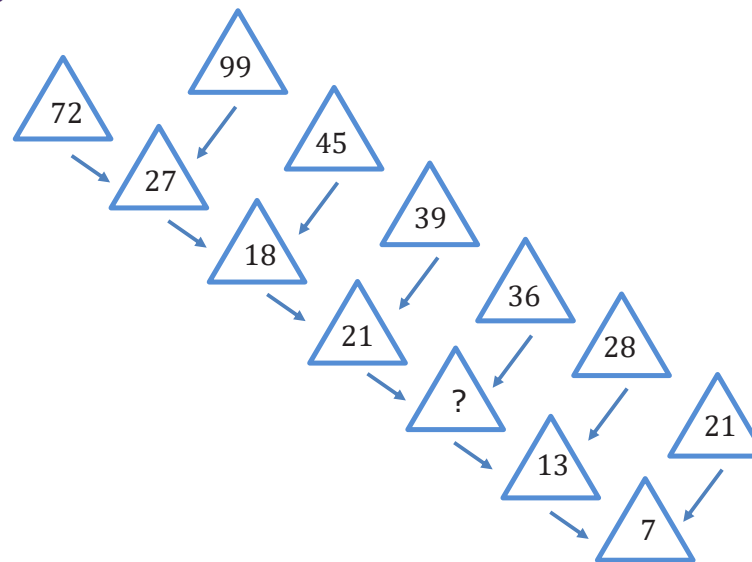


Learning Target

- ☐ I can apply properties of operations to generate equivalent expressions.



13.4.1 Warm-Up: Missing Number



1. What is the first number you tried for the missing number?
2. Did your first number work for the rest of the pattern?
3. What strategy did you try if your first number did not work?



13.4.2 Guided Instruction: Generating Equivalent Algebraic Expressions

1. Consider the following expressions.

$$8 \cdot x + 8 \cdot 9$$

$$8(x + 9)$$

$$8x + 9$$

- a. Which of these expressions are equivalent?
- b. Which property of operation justifies that the expressions are equivalent?

2. Write two equivalent expressions. Name the property or properties you used to create each equivalent expression.

3. For each expression, use the associative and commutative properties to generate an equivalent expression.

Original Expression	Associative Property	Commutative Property
$(3x + 5) + 4z$		
$8 \cdot (-1 \cdot 13d)$		

4. Rewrite $-11b + 3(10k + 1)$ using the distributive property.
5. Rewrite $-11b + 3(10k + 1)$ using the commutative property of multiplication.
6. Rewrite $-11b + 3(10k + 1)$ using the commutative property of addition.

7. Makayla and Soledad both rewrote $(-5h + 9) \cdot -14$ using the commutative property of multiplication. Verify that their expressions are equivalent by substituting a value in for h . Show your work in the table.

Makayla	Soledad
$-14(-5h + 9)$	$(h \cdot -5 + 9) \cdot -14$
$h = \underline{\hspace{2cm}}$	$h = \underline{\hspace{2cm}}$



13.4.3 Try It: Writing Equivalent Algebraic Expressions

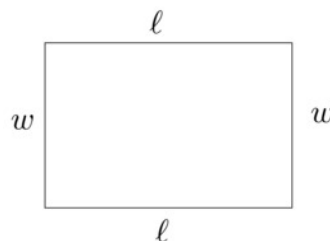
- Select the expressions that are equivalent to $(2x + 7) + (5y - 3)$.
☐ $(5y - 3) + (7 + 2x)$
☐ $(3 - 5y) + (7 + 2x)$
☐ $7 + (5y + 2x) - 3$
☐ $(7 + 5y) + (2x - 3)$
☐ $5y + (7 + 2x) - 3$
- Write two equivalent algebraic expressions for each expression below.
 - $(4 + 9) \cdot x$
 - $3t \cdot 4$
 - $-7x + 3 - 14$
 - $3(-9p - 4) + 7$



13.4.4 Your Turn!

- The students in Ms. Justino's class are writing expressions for the perimeter of a rectangle of side length, ℓ , and width, w . The expressions for four students are in the table.

Student	Expression
De'Andre	$2(\ell + w)$
Summer	$\ell + w + \ell + w$
Abm	$2\ell + w$
Luzarie	$2\ell + 2w$



Ms. Justino told the group that one of the expressions is not equivalent. Write a short explanation to the student to show why their expression is not equivalent to the other expressions.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

- Match equivalent expressions by drawing an arrow. If an expression does not have a match, write an equivalent expression to match it in one of the blank boxes provided.

Column A
$-8(x + 3)$
$-8x \cdot 3$
$(8x \cdot -3) \cdot 2$
$(3 + -8x) + 2$
$2 + -8x$

Column B
$2 + (-8x + 3)$
$3 \cdot -8x$
$-24 - 8x$



13.4.5 Wrap-Up: Error Analysis

Tangela was asked to circle all of the expressions equivalent to $6(y + 1)$.

Her work is shown below.

$6y + 1$	$6y + 7$	$6(y) + 1(y)$	$6(y) + 6(1)$	$6y + 6$
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1. Determine if Tangela correctly identified all of the equivalent expressions. If she is missing an expression, circle the expression(s) she is missing.
2. If Tangela made an error, then explain her error and list the equivalent expressions.

Unit 13, Lesson 5: Writing Algebraic Expressions



Benchmark

MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.



Learning Targets

- ☐ I can translate a written description into an algebraic expression.
- ☐ I can translate a real-world description into an algebraic expression.